US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

*Test Operations Procedure 03-2-501 14 May 2026

WATER GUNNERY FOR AMPHIBIOUS VEHICLES REAL FIRING FIELD TESTS

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1. SCOPE.

The procedures in this Test Operations Procedure (TOP) describe the conduct and procedures to conduct combat vehicle water gunnery operations. These procedures will include planning conduct and analysis of hit probability of a complete amphibious combat vehicle system operating in the water against well-defined reference targets either on land or water. Tests are conducted under a variety of measurable and, if possible, reproducible conditions (e.g., combinations of modes, ranges, ammunition, courses, vehicles, and targets).

2. FACILITIES AND INSTRUMENTATION.

The instrumented firing range includes the measuring of equipment required for obtaining all needed parameter data (inside and outside the vehicle), controlling the "firing quality" to ensure both the correct evaluation of firing performance, and the explanation or interpretation of data collected.

2.1 Facilities.

2.1.1 Test Area/Facility.

a. Range and Impact Area. A land area where targets can be positioned at varying distances that is large enough to accommodate the surface danger zone of the combat vehicle mounted weapon system under test.

b. Navigable Waterway. Conduct of water gunnery necessitates access to navigable waterways controllable for safety and must be operationally relevant to the system under test. This waterway must support the full spectrum of potential operating environments ranging from calm, littoral areas, including beaches and inland bays, to deeper open water conditions such as the Gulf of America, Atlantic Ocean, or Pacific Ocean.

2.1.2 Specialized Equipment.

a. Grid Board. A panel with grid spacing marked to permit the location of aiming points to within 0.1 mrad.

(1) A calibrated grid board in a vertical orientation, plumb, level, and perpendicular to the line-of-sight (LOS) at the distance from the weapon that corresponds to the gridboard calibration. The gridboard should be placed at a known distance from the vehicle location, with indicators or marks with spacing that correlates to an angular measurement. An example of a 500-m gridboard is shown in figure 1.

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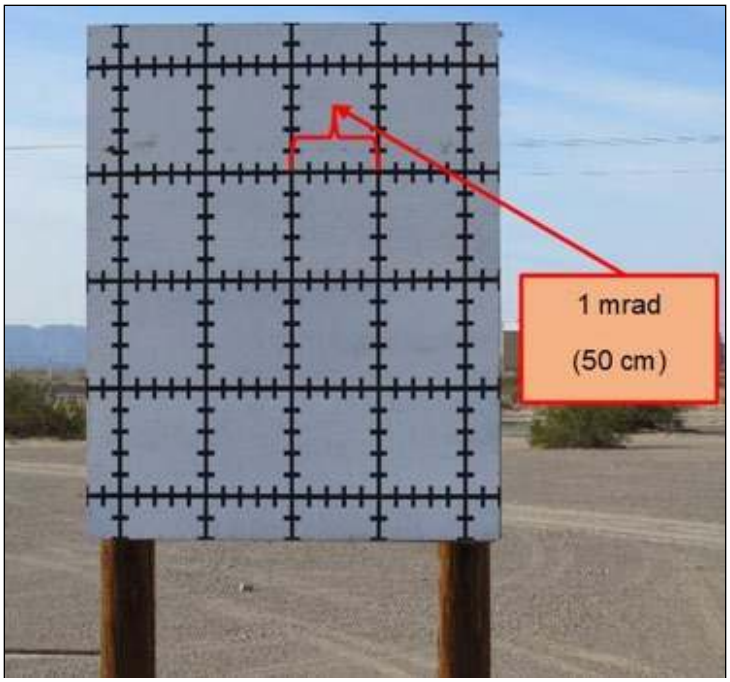


Figure 1. Example of a 500-m target gridboard.

(2) Given the desired distance between the vehicle and the gridboard location, and the desired resolution for angular measurements, the required indicator/mark spacing can be calculated by:

$$S = r\theta$$

where:

S = indicator/mark spacing (i.e., arc length) = 50 cm.
r = distance between vehicle and grid board = 500 m.
Θ = angular increment (i.e., central angle) = 1 mrad.

It is important that both the day sighting system and any thermal or night vision sighting systems are tested. A separate gridboard can be used, or the primary gridboard can be marked with a means of viewing the measurement increments with a thermal/night vision system so that the alignment can also be tested.

b. Boresight Device/Alignment Tool. The type of fire control system (FCS), physical limitations of test hardware, instrumentation, and objectives of the test determine what type or types of bore-alignment references should be used. Boresight devices can be small, medium, or large caliber depending on the system under test as shown in figure 2.

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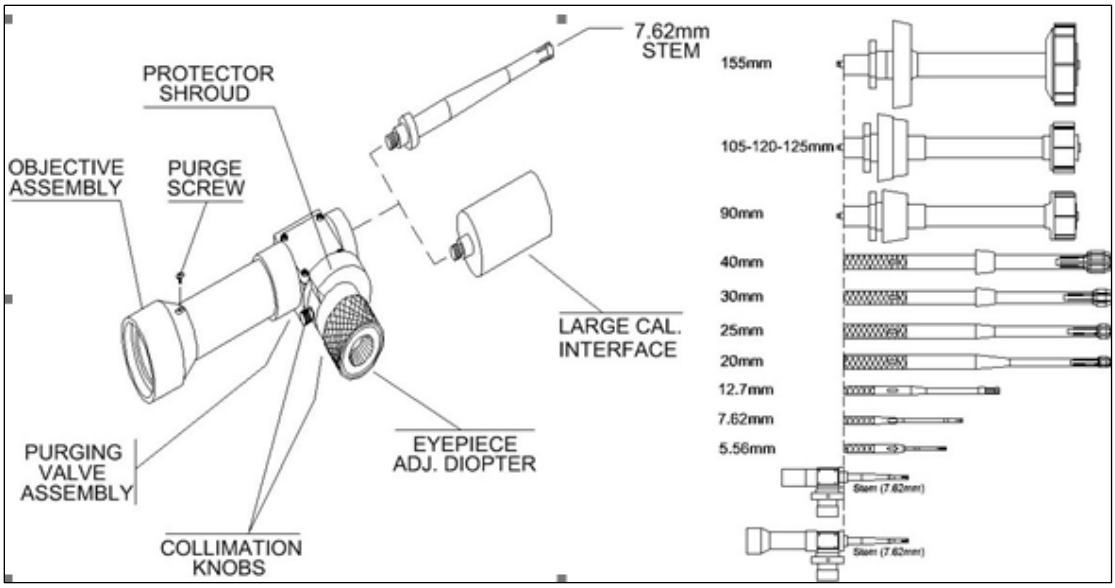


Figure 2. Boresight devices.

c. FCS Data Bus. Additional data acquisition (DAQ) equipment can be used to collect data directly from FCS data buses and digital communications lines that contain information regarding operational modes, system faults, engineering data, raw sensor data, external data communications, and system control signals. Typically, documentation (i.e., Interface Control Document) from the system developer is necessary to determine the protocol and message structure of each data source to be processed. Examples of other useful FCS data are sensor signals, ballistic computer inputs/outputs, system-perceived environmental conditions, and vehicle position/orientation. It is important to note that developer-based data acquired from vehicles can be useful as supplementary information to the primary instrumented data. It should not be used in lieu of verified instrumentation methods of DAQ unless signal accuracy is verified before testing.

d. Targets. Various types of targets (Fig. 3) can be used from simple aim points to operationally realistic silhouettes depending on the scope and methodology of the test.

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Figure 3. Silouette and aimpoint targets.

(1) Land target. Targets can be stationary or moving on land and will be fired on from the combat vehicle on the water. The moving target system and scoring system should be protected from incoming fire to the extent possible. An example is shown in Figure 4.



Figure 4. Example of land targets.

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(2) Water target. Targets can be stationary to the extent possible or moving on the water and will be fired on from the combat vehicle also on the water. An example of the U.S. Army Aberdeen Test Center (ATC), Aberdeen Proving Ground (APG), Maryland, Scorable Maritime Towed Target (SMarTT) (app B) is shown in Figure 5 and its remote-controlled tow boat in Figure 6.



Figure 5. SMarTT.



Figure 6. Remote-controlled tow boat.

(3) Air target. Targets can be drones (Fig. 7) or aircraft flying following constant velocity or evasive paths and fired on from the water.

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Figure 7. Aerial target drone.

e. The sighting system aim point and weapon pointing direction are observed and measured using cameras. A camera is mounted on a weapon surface so that the camera maintains alignment with the weapon (fig. 8). For sight reticle aim point observation, direct use of sighting system video is preferred. Alternatively, a camera may be mounted to a sighting system eyepiece. Position the target forward of the vehicle at a distance great enough to remain in the field-of-view of the weapon camera and sight video/camera throughout all required test runs.

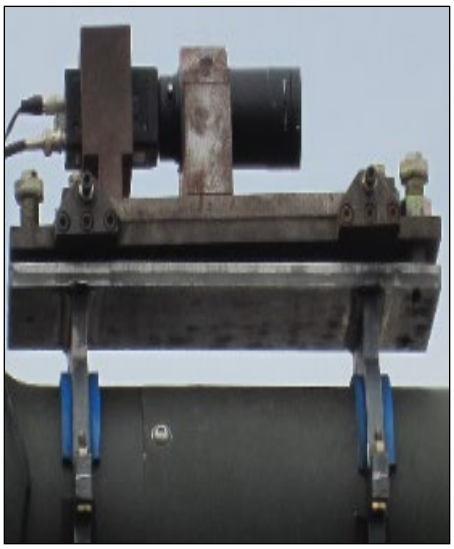


Figure 8. Example of a weapon camera.

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f. Inertial Measurement Unit (IMU). An IMU, usually consisting of triaxial microelectromechanical system gyros and accelerometers, may be used to determine weapon and sight pointing directions, drift characteristics, and dynamic angular rates. IMUs will be mounted on surfaces so that the devices maintain alignment with the weapon and sight (Fig. 9). The benefit of this method is that a reference target is not required (determination of static coincidence window only). Use of IMUs may be beneficial during dynamic testing, since after initial target alignment is achieved optical tracking of reference targets is not required. For some sighting systems, this method will not capture the gunner sight reticle aim point due to inaccessibility of internal sighting mechanisms such as mirrors or gimbals.



Figure 9. Example of an IMU.

2.2 Instrumentation. The instrumentation required to conduct this test is outlined in table 1.

TABLE 1. INSTRUMENTATION

Devices for Measuring	Permissible Measurement Error
Test vehicle location	1 m
Target location	1 m
Projectile impact location	0.05 mrad
Sight LOS to target aim point	0.05 mrad
Gun LOS to target aim point	0.05 mrad
Test vehicle speed	0.1 mph
Target speed	0.1 mph
Ammunition temperature	1 °F
Ambient air temperature	0.1 °F

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TABLE 1 (CONT)

Devices for Measuring	Permissible Measurement Uncertainty
Barometric pressure	0.4 in./H_2O
Wind speed (mean)	0.1 mph
Wind direction	1 deg
Elapsed time	0.1 s
Sea state	Observed
Water current (speed)	1 mph
Angular rate of hull (roll, pitch, yaw)	0.5 mrad/s
Angular rate of weapon (roll, pitch, yaw)	0.5 mrad/s

Note: All instrumentation calibration must be valid through the end of the current test and traceable to the appropriate national or international standard.

3. REQUIRED TEST CONDITIONS.

3.1 Inspection and Servicing.

a. The vehicle should be pre-mission preventive maintenance checks and service in accordance with applicable Technical Manuals (TMs). All vehicle subsystems should be verified to be in working order with no faults or deficiencies that could impact safety or performance of the system.

b. The vehicle should be equipped with support equipment or surrogate weight to reflect the operationally relevant weight, center of gravity (CG), and buoyancy of a vehicle equipped for combat.

c. Ammunition and any equipment required to execute testing, such as radio batteries, should be verified to be in good condition and of an appropriate quantity to conduct the full test.

d. Any onboard safety equipment, such as personal flotation devices, compressed air bottles, and other equipment should be verified as it is present, serviceable, and secure. Specific requirements for what safety equipment is required to conduct operations should be determined by tester, local and Army safety policy.

3.2 Controlled Test Conditions. Some variables outlined in the test plan can be controlled and should be verified to be within the requirements outlined in the test plan. These include test location, time of day, and vehicle configuration.

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3.3 Uncontrolled Test Conditions. As required, some variables are outside the tester's control. If the variable is a critical test requirement, such as sea state, or wind speed/direction, the tolerance to the test requirement should be assessed by the tester and determination can be made if the test can continue. If the critical but uncontrollable variable is outside the required tolerance, testing should be delayed and conditions should be monitored. All test conditions regardless of controllability should be recorded. This would include variables such as water current speed, wind speed, and direction when within allowable limits.

3.4 Test Matrix. A table of desired test variables should be developed to reflect the intent of the water gunnery test. This matrix should be developed in accordance with test objectives, programmatic objectives, and an overall evaluation plan. A simplified test matrix is presented in table 2 and a full matrix is available in appendix A.

TABLE 2: SAMPLE WATER GUNNERY TEST MATRIX

Vehicle		Target		Range to Target (m)	Sight	Ammunition Type
Speed (mph)	Direction	Speed (mph)	Direction			
3	Left to right	0	NA	750	Gunner	AP
3	Right to left	0	NA	750	Gunner	AP
3	NA	10	Left to right	850	Auxiliary or emergency	AP
3	NA	10	Right to left	850	Commander	HE
10	Left to right	11	Right to left	1000	Gunner	AP
10	Right to left	11	Left to right	1000	Gunner	HE
15	Left to right	15	Left to right	1500	Commander	HE
15	Right to left	15	Right to left	1500	Gunner	AP
28	Closing	10	Left to right	[a]2000	Gunner	AP
28	Closing	10	Right to left	[a]2000	FLIR	HE
3	Closing	0	NA	[a]3500	Gunner	ATGM

[a]Initial distance at start of engagement.

AP = armor piercing
ATGM = Anti-tank Guided Missile
FLIR = forward-looking infrared
HE = high explosive
NA = not applicable

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4. TEST PROCEDURES.

4.1 Test Interval Definitions.

a. Scenario. Test scenarios are groupings of vehicle and target movements and target range conditions. Movements include stationary vehicle-stationary target (S-S), stationary vehicle-moving target (S-M), moving vehicle-stationary target (M-S), and moving vehicle-moving target (M-M) conditions. Moving vehicle scenarios may be further divided the direction and path of travel of the vehicle toward, oblique, or parallel to the target. Moving target scenarios may include constant velocity, maneuvering, and evasive (accelerating and closing range) targets. Scenario conditions will significantly impact stabilization error, a random error, and ultimately hit probability.

b. Occasion. A firing occasion includes all rounds fired from the same combat vehicle during a period in which no significant events (except firing the weapon) occurred that could affect FCS, weapon system, or ammunition delivery accuracy characteristics. Occasions occur within scenarios.

Further, occasions are test instances of a particular combat vehicle or ammunition type where the set of variable biases and associated error sources are held constant, as best possible. Test configuration and timing changes that affect variable biases constitute a change in occasion. Changes include vehicle sample, gun tube sample, vehicle cant condition, ammunition temperature, target range, target speed, test course, test day (usually due to changes in environmental conditions), and significant changes in air temperature and steady wind speed/direction within a test day (requiring some tester judgement). For each test project, it is recommended to define the parameters of an occasion in detail to ensure that both field testers and data analysts/statisticians agree on grouping of resulting datasets.

c. Groups. Groups are sequenced sets of firing events within occasions. For example, a large caliber firing occasion may require 20 rounds of a particular ammunition to be fired at a constant speed target, but the rounds are fired in four five-round groups. Practical reasons for groups include time available to fire on a moving target, firing clearance limitations, ammunition and loading limitations, weapon and target maintenance, firing cadence limitations, and personnel breaks. The terms “groups” and “occasions” may be synonymous if an occasion consists of a single group.

d. Bursts. Bursts are multiple round engagements within groups or occasions fired from automatic or chain gun weapon systems.

e. Engagements. Engagements refer to individual acts of laying on a target and firing a single shot or burst.

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f. System Software (SW) Configuration Changes. Firing intervals do not include FCS software configuration changes. SW configuration changes may affect stabilization error, ballistic computation errors, ranging errors, and many other fixed and variable biases and random error sources. Since the effects of SW configuration changes on fire control performance are difficult to determine by test personnel, data before and after software configuration changes should be treated as distinct samples of data during analysis.

4.2 Test Procedures.

4.2.1 Pretest –Before Water Entry.

a. Before testing, a check for function and inspection should be conducted on all instrumentation to verify it is calibrated and functional.

b. The sighting and weapon systems that will be tested should be boresight in accordance with appropriate TMs or procedures. Boresight values input into the FCS should be recorded.

c. Verify all safety equipment, support equipment and ammunition are loaded onto the test vehicle to minimize the need to resupply while in the water.

4.2.2 Pretest–After Entry to Water.

a. Any component that requires warm-up, such as hydraulics or electrical systems, should be allowed to come up to operating temperature before the start of live fire testing and should be done in accordance with appropriate TMs.

b. If required, a “dry run” of the first test scenario should be run to verify the timing of moving targets, test vehicle, instrumentation data collection, photography equipment and tester control of the engagement works as intended. It is recommended that any remote-controlled systems of targetry are also exercised during this time to verify function, communication, and scoring ability.

4.2.3 Zeroing.

a. After boresighting, the weapon system under test can be zeroed using a reference target or test target as specified by the TM and test plan. Zero adjustment value should be recorded.

b. If test extends for long periods of time, if results begin to shift in a direction, and if the TM/test plan allows, re-zeroing can be conducted when called for. Any change in zero numbers into the FCS should be recorded.

4.3 Live Fire Water Gunnery Testing.

4.3.1 Test Matrix Execution. A tabular table of desired variables should be followed to ensure that all relevant parameters are tested, and a statistically significant number of rounds is fired to

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provide a basis for data presentation. A larger sample of a test matrix is included in appendix A. This test scenario includes multiple weapons, weapon sight systems, sight settings, weapon control station, and sight magnification settings, but should not be considered an all-inclusive table. The test matrix should be specifically tailored to meet the tester's or evaluator's data requirements.

4.3.2 Document Changes. As testing is conducted, take care to note any faults, cautions, or warnings that may appear on vehicle systems during testing. Testing should continue if faults do not interfere with system performance. If testing is conducted over longer periods of time where system changes may occur, ensure to note any changes in major or subsystem SW versions, weapon component changes, along with any parts changed out on the test vehicle that could add variability to test results. As testing is conducted in the water, salt spray, and residue can build up on optics and view ports. When appropriate and in accordance with the test plan, operators should ensure they have a clear view of the path of travel of the vehicle, along with general situational awareness.

4.3.3 Degraded Modes. If applicable, testing can be conducted with the FCS in emergency modes or manual modes to intentionally test the performance of the system in these modes.

5. DATA REQUIRED.

a. Projectile impact location (horizontal and vertical coordinates).

b. Laser range finder (LRF) distance to target, fire control input.

c. Vehicle-to-target range, LRF input to FCS.

d. Sight LOS referenced to target aimpoint (azimuth and elevation).

e. Gun LOS referenced to target aim point (azimuth and elevation).

f. Gun angular rates (pitch, roll, and yaw).

g. Vehicle angular rates (pitch, roll, and yaw).

h. Target speed and direction.

i. Vehicle speed and direction.

j. Meteorological conditions (air temperature and pressure, wind speed and direction, sea state).

k. Common time base for all events.

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l. Boresight values.

m. Zero values.

n. FCS data bus system.

o. FCS firing rate settings (single shot, burst, and automatic).

p. Sight settings and magnification (x3, x10, filter, and clear).

q. Vehicle position, at time of firing.

r. Target position, at time of firing.

6. PRESENTATION OF DATA.

6.1 Pre-Firing Data. Report the data required in paragraph 5 in paragraph or tabular format, as appropriate.

6.2 Firing Event Data. Report the following data in tabular or graphical format, as appropriate.

a. Firing dates, times, number of rounds, range location, and line-of-fire (from north or south).

b. Meteorological data, including air temperature, barometric pressure, relative humidity, wind speed, and wind direction from range instrumentation.

c. FCS data bus signal recordings (optional). Relevant signals may be reported in graphical format, as appropriate, when used to explain test result anomalies or for comparison purposes.

d. High-speed camera images for ammunition behavior (optional). Images may be presented in graphical format to demonstrate observed anomalies or for comparison purposes.

e. Actual distance to target. Difference between LRF range and actual range to target (para 5p and 5r).

6.3 Post-Firing Event Data. Report data requiring post-processing of firing event data. Data are generally reported in tabular or graphical format (e.g., azimuth and elevation impact plots), as appropriate. At a minimum, report the following for each burst (medium-caliber, automatic weapon only) and scenario set of firings.

a. Hit Ratio. Using the total number of rounds fired compared to the total number of target impacts the hit ratio can be expressed for each burst, engagement, scenario, or group. An example is presented in tables 3 and 4.

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TABLE 3. HIT RATIO PER BURST, SCENARIO 1

Burst No.	Total		Hit Ratio
	Hit	Fired	
1	6	10	0.60
2	5	10	0.50
3	4	11	0.36
4	5	10	0.50
5	6	9	0.67

TABLE 4. HIT RATIO PER SCENARIO

Scenario No.	Total		Hit Ratio
	Hit	Fired	
1	322	500	0.64
2	299	500	0.60
3	307	495	0.62
4	411	500	0.82
5	133	425	0.31
6	2	3	0.67

b. Miss Distance for Each Dataset. The X and Y axis impact data for each test can be plotted and expressed in relation to the target size, or requirement if a virtual target is used. A sample data plot of a single ammunition type in various vehicle/target conditions is shown to be overlayed on its target requirement in Figure 10.

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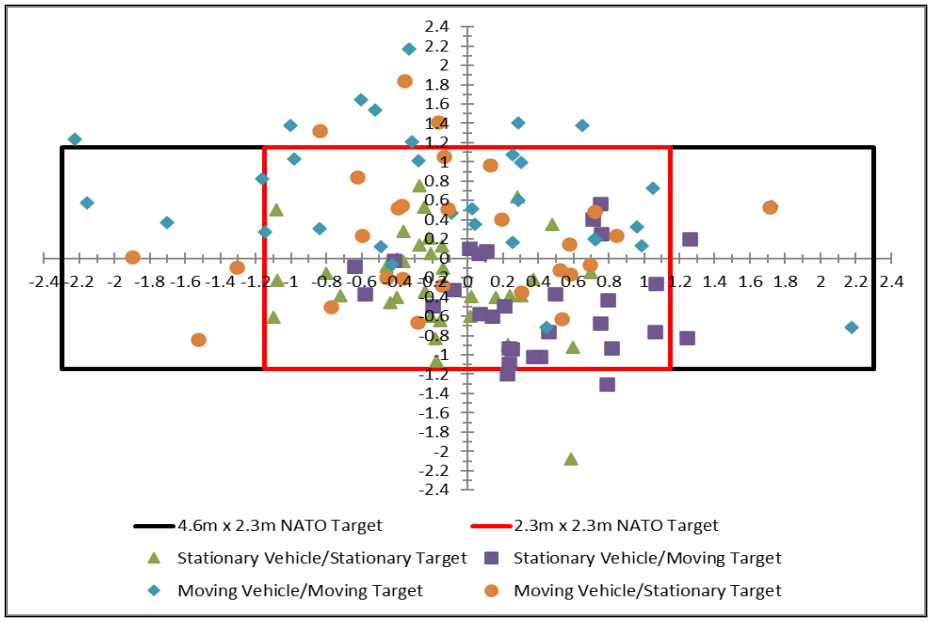


Figure 10. X and Y plot of impacts overlayed on target requirement.

c. Aim Error. Report aim error mean and standard deviation values for each burst and firing occasion. These results may help quantify boresight error characteristics and will allow data processing to remove gunner aim error from the impact data, if required. Aim error is expected to be small during Stationary-Stationary Ammunition Accuracy Tests (SSAATs) for single-shot events or the first round in a burst event.

d. FCS Sensor Errors. If FCS sensor (e.g., FCS wind, range, ambient temperature, and barometric pressure sensors) outputs are compared to independent sensors during testing, report errors in tabular format, including means and standard deviations.

e. Calculate Hit Probability and Describe Method Used.

(1) Single-shot hit probability. Report single-shot hit probability for all ranges required, taking care to comply with all security classification guidelines. Ammunition type, range, dispersion, and hit probability results will not be combined and displayed in an unclassified setting or document.

(2) Probability of K hits from an N round burst. For medium-caliber, automatic weapon systems report the probability of K hits from an N round burst, taking care to comply with all security classification guidelines. Ammunition type, range, dispersion, and hit probability results will not be combined and displayed in an unclassified setting or document.

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(3) Expected number of hits for a given burst of N rounds. For medium-caliber, automatic weapon systems report the expected number of hits for a given burst of N rounds, taking care to comply with all security classification guidelines. Ammunition type, range, dispersion, and hit probability results shall not be combined and displayed in an unclassified setting or document.

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APPENDIX A. EXAMPLES OF EXPERIMENTAL DESIGNS

TABLE A-1. FIRING TEST MATRIX

Vehicle		Target		Range to Target (m)	Target Location	Sight	Control	Ammunition Type	Zoom	Weapon	Fire Control Setting
Speed (mph)	Direction	Speed (mph)	Direction								
0	NA	0	NA	500	Land	Day	Gunner	AP	x3	Main	Single
0	NA	0	NA	500	Land	Day	Gunner	AP	x3	Main	Burst
0	NA	0	NA	500	Land	FLIR	Gunner	AP	x3	Main	Auto
0	NA	0	NA	500	Land	FLIR	Gunner	AP	x3	COAX	Auto
0	NA	0	NA	500	Land	Day	Commander	AP	x3	RWS	Auto
0	NA	0	NA	500	Land	Auxiliary	Gunner	AP	Auxiliary	Main	Auto
5	Left to right	10	Left to right	650	Land	FLIR	Commander	HE	x3	Main	Burst
5	Right to left	10	Right to left	650	Land	FLIR	Commander	HE	x10	Main	Single
3	Left to right	0	NA	750	Land	Day	Gunner	AP	x3	Main	Auto
3	Right to left	0	NA	750	Land	Day	Gunner	AP	x3	Main	Auto
3	Closing	10	Closing	[a] 500	Water	Auxiliary	Auxiliary	HE	x3	Main	Auto
3	Closing	10	Closing	[a] 850	Water	FLIR	Commander	HE	x3	Main	Auto
10	Left to right	11	Right to left	1000	Water	Day	Gunner	AP	x3	Main	Auto
10	Right to left	11	Left to right	1000	Water	Day	Gunner	HE	x3	Main	Auto
15	Left to right	15	Left to right	1500	Water	Day	Commander	HE	x10	Main	Auto
15	Right to left	15	Right to left	1500	Water	FLIR	Gunner	AP	x3	Main	Auto
15	Closing	Variable	Evasive	[a]1500	Air	Day	Commander	HE	x3	Main	Auto
15	Closing	Variable	Evasive	[a]1500	Air	Day	Gunner	HE	x10	Main	Auto
28	Closing	10	Left to right	[a]2000	Land	FLIR	Gunner	AP	x10	Main	Auto
28	Closing	10	Right to Left	[a]2000	Land	FLIR	Gunner	AP	x10	Main	Auto
3	Closing	0	NA	3500	Land	Day	Commander	JAV	x50	RWS	ATGM
3	Closing	0	NA	3500	Land	FLIR	Gunner	ATGM	x25	ATGM	ATGM

[a]Initial distance at start of engagement.

JAV = Javelin ATGM = Anti-Tank Guided Missile
RWS = Remote Weapon Station

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APPENDIX B. SCORABLE MARITIME TOWED TARGET (SMarTT)

The SMarTT target is a real-time scorable target system mounted on a pontoon boat, which is pulled behind a remotely operated high-speed maneuverable surface target (HSMST) (fig. B-1). This target gives real-time hit information to the users.



Figure B-1. Target towed behind a robotic boat.

a. Provides real-time hit sensing.

b. Can be pulled up to speeds of 30 mph depending on sea state.

c. The system is easily transported to any range complex needed and is based out of the ATC, APG, MD.

d. This system has been used in Hawaii at waterborne ranges 40 mi off the coast, and off Virginia Beach to support Army and USMC maritime gunnery training with great success.

e. The exterior can be fitted with thermal pads to give a heat signature.

f. Black panels are scorable. Top panels are made for top-down firing from a helicopter (Fig. B-2).

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Figure B-2. Top panels are made for top-down firing from a helicopter.

g. Waypoint mode navigation of the remote tow boat interface (Fig. B–3).



Figure B–3. Waypoint mode navigation of the remote tow boat interface.

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h. Supporting U.S. Army training off Hawaii (Fig. B-4).



Figure B–4. Offshore training support.

i. At completion of a gunnery event off Hawaii, with 2 hr of repairs, this target was ready to be used again (Fig. B-5).



Figure B-5. Post-training damage.

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APPENDIX C. ABBREVIATIONS

AP	armor piercing
APG	Aberdeen Proving Ground
ATC	U.S. Army Aberdeen Test Center
ATGM	Anti-Tank Guided Missile
CG	center of gravity
DAQ	data acquisition
FCS	fire control system
FLIR	forward-looking infrared
HE	high explosive
HSMST	high-speed maneuverable surface target
IMU	inertial measurement unit
JAV	Javelin
LOS	line-of-sight
LRF	laser range finder
M-M	moving vehicle-moving target
M-S	moving vehicle-stationary target
RWS	remote weapon station
SDZ	surface danger zone
S-M	stationary vehicle-moving target
SMarTT	Scorable Maritime Towed Target
S-S	stationary vehicle-stationary target
SSAAT	Stationary–Stationary Ammunition Accuracy Test
SW	software
TM	technical manual
TOP	test operations procedure

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APPENDIX D. APPROVAL AUTHORITY

DEPARTMENT OF THE ARMY
UNITED STATES ARMY TEST AND EVALUATION COMMAND
6617 ABERDEEN BOULEVARD, BUILDING 2202–THIRD FLOOR
ABERDEEN PROVING GROUND, MD 21005-5001

TEWS-CG (73-1f)

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Test Operations Procedure 03-2-501 Water Gunnery for Amphibious Vehicles Real Firing Field Tests Publication

1. Test Operations Procedure (TOP) 03-2-501 Water Gunnery for Amphibious Vehicles Real Firing Field Tests has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Evaluation Command, and the U.S. Army Public Health Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This Test Operations Procedure (TOP) describes the conduct and procedures to conduct combat vehicle water gunnery operations.

3. This document is approved for publication and will be posted to the ASSIST website. The ASSIST website can be accessed at https://assist.dla.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army White Sand Missile Range (TEWS-CG), 100 Headquarters Avenue, White Sands Missile Range, NM 88002-5000; or e-mailed to usarmy.apg.atec.mbx.atec-standards@army.mil.

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